COST ESTIMATOR AI AGENT

Enterprise Implementation Plan

Enterprise Stack · SOP-Aligned · Validation-First Deployment

Start Date: 25-Mar-2026 | Target Go-Live: 03-May-2026

Phase	Focus	Dates	Milestone
1	Validation & Accuracy Benchmarking	25-Mar-2026 to 03-Apr-2026	Accuracy scorecard approved
2	Live Manual Input - Single-Product	04-Apr-2026 to 10-Apr-2026	Single-product costing live
3	Quote Automation - BOQ & Multi-Item	11-Apr-2026 to 18-Apr-2026	Multi-item BOQ processing
4	EPC-Level Estimation - Welding, NDT	19-Apr-2026 to 25-Apr-2026	Full EPC estimate + compliance
5	Enterprise Scaling - Multi-Agent	26-Apr-2026 to 03-May-2026	Enterprise platform deployed

1. Enterprise Stack Architecture

The Cost Estimator Agent is built on a modular four-layer enterprise stack, ensuring scalability, governance, and full auditability. Every component is selected for self-hosted control, cost efficiency, and compatibility with the broader enterprise AI platform.

Architecture Flow: User → Orchestrator → Agent(s) → Tools → Memory → LLM → Output

1.1 Orchestrator Layer

Component	Details
Technology	n8n — Self-Hosted Workflow Automation Platform
Why n8n	Open-source, zero per-execution cost, visual workflow builder, 400+ native integrations, built-in AI agent nodes, full data sovereignty
Responsibilities	Route requests to correct agents via webhook triggers; manage tool invocation sequence; handle approval workflows; maintain full audit logs to PostgreSQL

1.2 LLM Layer

Component	Details
Technology	Azure OpenAI — GPT-4o (Multimodal: text + vision)
Capabilities	Engineering drawing interpretation (vision), RFQ text extraction, weight and cost calculations, structured output generation, flagging missing/ambiguous data
Why GPT-4o	Single model handles both text reasoning and drawing vision — eliminates need for a separate OCR layer for well-structured PDF drawings

1.3 Memory Layer

Layer	Technology	Stores
Raw Storage	Azure Blob Storage	RFQs, drawings, BOQs, costing sheets, quotations
Structured Data	PostgreSQL	Quote history, user logs, approvals, roles, rate library
Vector Memory	pgvector Extension	Semantic search across all documents — similar-job retrieval
Session Memory	n8n Workflow Context	Active job state — current RFQ, extracted data, intermediate values

1.4 Tools Layer

#	Tool	What It Does	Output
1	Drawing Reader	Extracts dimensions from PDF drawings using OCR + GPT-4o vision. Identifies plates, pipes, beams, channels.	Structured dimension object
2	Weight Calculator	Calculates net weight, gross weight (with scrap factor) using geometry formulas for standard steel sections.	Weight breakdown (net/gross/scrap)
3	Excel Costing Sheet Generator	Populates standard Excel costing template with material, labour, overhead, and totals. Preserves all formulas.	Completed .xlsx costing sheet

4	Quote Summary Generator	Creates concise summary with weight breakdown, cost breakdown, and key assumptions for review.	Quote summary report
5	BOQ Parser	Extracts line items from Bill of Quantities documents for multi-item costing across complex jobs.	Structured line-item list
6	Covering Letter Generator	Customizes covering letter using Master Fabrication Template, selecting clauses to project scope.	Covering letter draft

⚠ *Agent Split Policy: If any agent is assigned more than 6–10 tools, it must be reviewed for decomposition into smaller specialized agents. Current tool count: Cost Estimator (4), Quote Generator (2), Compliance Agent (3) — all within limit.*

1.5 Agent Layer — Sub-Agent Design

Agent	Tools	Function
Cost Estimator Agent	4	Interpret RFQ, extract dimensions from drawing, calculate weight, generate costing sheet and summary
Quote Generator Agent	2	Produce formatted techno-commercial offers and cover letters from finalized costing data
Compliance Agent	3	Validate outputs against welding/NDT specifications and material certifications (WPS, MTC)

2. Phase 1 — Validation & Accuracy Benchmarking

Dates: 25-Mar-2026 to 03-Apr-2026

Design Principle: Phase 1 is a controlled testing environment. The agent processes historical jobs where the correct output (manually prepared costing sheet) is already known. Every agent output is compared line-by-line against the manual estimate. Only after achieving defined accuracy thresholds does the team proceed to Phase 2 live implementation.

2.1 Validation Objectives

- Prove the agent can interpret engineering drawings and extract dimensions with ≥90% accuracy
- Verify weight calculations (net, gross, scrap) match manual calculations within ±5% tolerance
- Confirm costing sheet output matches the approved manual costing sheet structure and values
- Identify systematic errors, missing rules, or unsupported product types before going live
- Build estimator confidence in the agent by demonstrating output quality on real past jobs
- Establish a baseline accuracy score that will be tracked as the system improves

2.2 Historical Quote Selection Criteria

Criterion	Details
Product Variety	Plates, pipes, beams, channels, mixed assemblies — at least 2–3 quotes per product type
Complexity Range	Simple single-product quotes AND multi-item BOQ jobs to test both scenarios
Recency	Quotes from the last 12–18 months preferred (current rate structures and templates)

Complete Data	Only select quotes where the drawing, RFQ/BOQ, and final approved costing sheet are all available
Minimum Count	Minimum 10–15 historical quotes for a statistically meaningful validation set
Known Edge Cases	Include 2–3 quotes that were tricky or had manual corrections — tests the agent's edge-case handling

2.3 Validation Process Flow

Step	Action	Detail
1	Select Historical Quotes	Estimation team selects 10–15 completed quotes with all source documents (drawing, RFQ, approved costing sheet).
2	Agent Processes Quote	For each quote: upload drawing + customer docs to agent. Agent extracts dimensions, calculates weight, generates costing sheet.
3	Side-by-Side Comparison	Agent output placed next to the original manual costing sheet. Line-by-line comparison of: dimensions, weights, material cost, labour cost, overhead, total price.
4	Gap Analysis	Document every deviation: incorrect dimension, wrong weight, missing line item, wrong rate applied, formula error. Classify as Critical / Major / Minor.
5	Root Cause & Fix	For each gap: identify root cause (prompt issue, missing rule, drawing interpretation error, template issue). Apply fix.
6	Re-Run & Re-Compare	After fixes applied, re-run the same quotes. Compare again. Repeat until accuracy thresholds are met.
7	Accuracy Scorecard	Final scorecard produced: % accuracy per metric, pass/fail per quote, overall readiness assessment. Shared with management.
8	Management Sign-Off	Management reviews scorecard and approves or rejects progression to Phase 2. No live deployment without sign-off.

⚠ Phase 2 Gate: The team MUST NOT proceed to Phase 2 live implementation until the accuracy scorecard is reviewed and signed off by management. Any metric below its target threshold must be addressed and re-tested.

2.4 Validation Deliverables

- Validation Quote Set: List of 10–15 selected historical quotes with source documents
- Comparison Report: Side-by-side output for each quote (agent vs. manual)
- Gap Analysis Log: Every deviation documented with severity and root cause
- Accuracy Scorecard: Final pass/fail per metric, overall readiness score
- Fixes Applied Log: Record of every prompt, rule, or logic change made during validation
- Management Sign-Off: Formal approval to proceed to Phase 2

2.5 Phase 1 Detailed Timeline

Days	Dates	Activities	Owner
Day 1–2	25-Mar-2026 to 26-Mar-2026	Azure OpenAI provisioned; n8n setup; PostgreSQL + Blob Storage configured; Drawing Reader + Weight Calculator tools built	Dev Team
Day 2–3	26-Mar-2026 to 27-Mar-2026	Excel costing template integrated; manual input interface (chat UI) connected to workflow	Dev Team

Day 3–4	27-Mar-2026 to 28-Mar-2026	Estimation team selects 10–15 historical quotes; source documents collected and organized	Estimation Team
Day 4–7	28-Mar-2026 to 31-Mar-2026	Run all selected historical quotes through the agent; generate agent costing sheets for each	Dev + Estimation
Day 7–8	31-Mar-2026 to 01-Apr-2026	Side-by-side comparison and gap analysis for every quote; classify deviations	Estimation Team
Day 8–9	01-Apr-2026 to 02-Apr-2026	Root cause fixes applied to agent prompts, rules, and logic; re-run failed quotes	Dev Team
Day 9–10	02-Apr-2026 to 03-Apr-2026	Final accuracy scorecard produced; management review and sign-off for Phase 2	Dev + Management

Phase 1 Milestone (03-Apr-2026): Accuracy scorecard approved by management. Agent validated on real historical data. Team confident in output quality before any live use.

3. Phase 2 — Live Manual Input Layer (Single-Product Costing)

Dates: 04-Apr-2026 to 10-Apr-2026

Design Principle: Phase 2 is not a stepping stone — it is a permanent capability. Estimators will continue to use manual input for bespoke or urgent jobs even after full automation is live.

3.1 Required Manual Inputs

Input	Format	Notes
Engineering Drawing	PDF upload	GA or fabrication drawing with all dimensions indicated. Agent uses GPT-4o vision to extract.
Customer Documents	PDF / Word	RFQ, BOQ, scope of work, technical specification. Agent parses and pre-fills cost structure.
Steel Price	Number (AED/kg)	Current market rate for the steel grade being quoted. Manually entered or pulled from rate library.
Labour Assumptions	Free text	E.g., ‘40 manhours/unit at 35 AED/hr’. Agent parses and applies.
Quantity	Integer	Number of units — used as cost multiplier.
Product Name	Free text	Labels the output costing sheet, quote summary, and all related files for traceability.

3.2 Phase 2 Process Flow

Step	Action	Detail
1	Document Ingestion	User uploads drawing + customer docs. Agent confirms files received.
2	Dimension Extraction	GPT-4o vision reads the PDF drawing and extracts all dimensions, member types, quantities.
3	Weight Calculation	Agent calculates net weight, gross weight (with scrap factor), and scrap weight per item.
4	Cost Logic Application	Material rates, labour hours, overheads, and margins applied per company standard.

5	Costing Sheet Generation	Excel template populated with full cost breakdown. All formulas preserved.
6	Output Delivery	Agent returns completed .xlsx costing sheet + quote summary with download links.

3.3 Phase 2 Detailed Timeline

Days	Dates	Activities	Deliverable
Day 11–12	04-Apr-2026 to 05-Apr-2026	Apply all Phase 1 validation fixes permanently; chat UI refined based on estimator feedback	Refined agent
Day 12–14	05-Apr-2026 to 07-Apr-2026	Multi-user login via Azure AD (Entra ID); role permissions enforced; audit logging enabled	Multi-user access
Day 14–17	07-Apr-2026 to 10-Apr-2026	Estimators begin using agent for real single-product incoming quotes; parallel manual checking for first 3–5 live jobs	Live single-product costing

Phase 2 Milestone (10-Apr-2026): Single-product costing live for real incoming quotes with multi-user access and audit logging.

4. Phase 3 — Quote Automation (BOQ & Multi-Item Costing)

Dates: 11-Apr-2026 to 18-Apr-2026

4.1 Phase 3 Detailed Timeline

Days	Dates	Activities	Deliverable
Day 18–20	11-Apr-2026 to 13-Apr-2026	BOQ Parser tool built and connected to Costing Engine	BOQ parsing live
Day 20–23	13-Apr-2026 to 16-Apr-2026	Multi-item costing logic added to Excel generator (row-per-item with totals); labour rate rules configured by job type	Multi-item costing
Day 23–25	16-Apr-2026 to 18-Apr-2026	Covering Letter Generator tool integrated with Master Template	Cover letter automation

4.2 Phase 3 Deliverables

- BOQ Parser — extracts line items from Bill of Quantities documents (PDF and Excel formats)
- Multi-item costing — consolidated sheet with per-item and total costs across all BOQ lines
- Labour estimation — rule-based calculations using complexity factors by item type
- Covering letter automation — customized letter using Master Fabrication Template, clause selection per project

Phase 3 Milestone (18-Apr-2026): Multi-item BOQ processing and covering letter automation live.

5. Phase 4 — EPC-Level Estimation (Welding, NDT & Compliance)

Dates: 19-Apr-2026 to 25-Apr-2026

5.1 Phase 4 Detailed Timeline

Days	Dates	Activities	Deliverable
Day 26–28	19-Apr-2026 to 21-Apr-2026	Welding inch-diameter calculator built and calibrated against historical EPC jobs	Welding calculations
Day 28–30	21-Apr-2026 to 23-Apr-2026	NDT costing rules configured per method (RT, UT, MT, PT) and coverage %	NDT costing
Day 30–32	23-Apr-2026 to 25-Apr-2026	Compliance Agent built — WPS validator, MTC checker, checklist generator; Agent split review conducted	Compliance validation

Phase 4 Milestone (25-Apr-2026): Full EPC estimate capability with welding, NDT, and compliance validation.

6. Phase 5 — Enterprise Scaling (Multi-Agent + Knowledge Retrieval)

Dates: 26-Apr-2026 to 03-May-2026

6.1 Phase 5 Detailed Timeline

Days	Dates	Activities	Deliverable
Day 33–35	26-Apr-2026 to 28-Apr-2026	pgvector ingestion pipeline built — batch import all historical documents; semantic retrieval tested	Vector memory live
Day 35–37	28-Apr-2026 to 30-Apr-2026	Multi-agent orchestration assembled: Estimator → Compliance → Quote Generator chained in single workflow	Multi-agent pipeline
Day 37–39	30-Apr-2026 to 02-May-2026	Gmail OAuth 2.0 integration — automatic RFQ email ingestion, attachment parsing	Email automation
Day 39–40	02-May-2026 to 03-May-2026	UAT conducted with estimation team; user training across all roles; management sign-off; go-live	Enterprise platform deployed

6.2 Phase 5 Deliverables

- Document indexing — all historical costing sheets, drawings, and RFQs ingested into pgvector
- Historical quote retrieval — semantic search: ‘find similar past job by product, material, dimensions’
- Multi-agent orchestration — Estimator → Compliance → Quote Generator chained in single workflow
- Gmail middleware — OAuth 2.0, automatic RFQ email ingestion, attachment parsing
- Full audit log and version control live across all outputs
- User training conducted across all roles before go-live

Phase 5 Milestone (03-May-2026): Enterprise platform fully deployed. Go-live with full multi-agent orchestration and knowledge retrieval.

7. Multi-User Access & Data Storage

7.1 Authentication

Component	Details
Method	Enterprise SSO via Azure Active Directory (Entra ID)
Principle	Role-based access control (RBAC). Every user action is tied to their user ID and logged with a timestamp.

7.2 User Roles & Permissions

Role	Permissions
Estimator	Create estimates, upload drawings & documents, generate costing sheets, view own job history
Engineer	Review estimates, validate dimensions and technical assumptions, approve technical sign-off
Sales	View completed quotes, download proposals and covering letters
Manager	Approve quotes above threshold, view all team outputs, override margins, access division-level reports
Administrator	Manage users, configure rate libraries, access audit logs, manage templates and overhead structure

8. Standard Agent Response Structure

Every agent response must follow this format consistently for auditability, traceability, and user confidence:

#	Section	Contents & Purpose
1	Prompt Received	Agent restates the user's request in plain language. Confirms understanding before processing.
2	Files Uploaded	Lists every uploaded document by filename and type. Flags any file that could not be read.
3	Assumptions Used	States all values applied: steel price, density, scrap factor, labour rate, overhead %. User can correct.
4	Extracted Data Summary	Summary of dimensions, quantities, member types from drawing/documents. User confirms accuracy.
5	Outputs Generated	Costing sheet (.xlsx) and quote summary as download links. Named with JobID and date.
6	Flags & Warnings	Missing data, ambiguous values, items needing human review. Agent never silently skips issues.
7	Confidence Indicators	High / Medium / Needs Review per parsed value. Low-confidence items highlighted.
8	Next Step Prompt	What action is required: confirm, correct a flag, upload missing doc, or submit for approval.

⚠ *Design Rule: The agent must never generate a final costing output without first receiving user confirmation of the extracted data and assumptions.*

9. Complete Implementation Roadmap with Dates

Total Estimated Rollout: 40 calendar days (approximately 6 weeks) from 25-Mar-2026 to 03-May-2026. Phase 1 is validation-only; live implementation begins at Phase 2.

Phase	Focus	Days	Start Date	End Date	Milestone
1	Validation & Accuracy Benchmarking	1–10	25-Mar-2026	03-Apr-2026	Accuracy scorecard approved; management sign-off
2	Live Manual Input — Single-Product Costing	11–17	04-Apr-2026	10-Apr-2026	Single-product costing live for real quotes
3	Quote Automation — BOQ & Multi-Item	18–25	11-Apr-2026	18-Apr-2026	Multi-item BOQ processing
4	EPC-Level Estimation — Welding, NDT & Compliance	26–32	19-Apr-2026	25-Apr-2026	Full EPC estimate with compliance
5	Enterprise Scaling — Multi-Agent + Memory	33–40	26-Apr-2026	03-May-2026	Enterprise platform fully deployed

10. Next Steps & Immediate Actions

#	Action Required	Owner	Target Date
1	Provision Azure OpenAI subscription with GPT-4o access	Management / IT	Before 25-Mar-2026
2	Provide standard Excel costing template to development team	Estimation Team	Before 25-Mar-2026
3	Select 10–15 historical quotes for Phase 1 validation (with all source documents)	Estimation Team	25-Mar-2026 to 27-Mar-2026
4	Provide sample drawings and historical costing sheets for testing	Estimation Team	25-Mar-2026 to 27-Mar-2026
5	Set up n8n on company infrastructure (self-hosted)	Dev Team	25-Mar-2026
6	Configure Azure Blob Storage and PostgreSQL database schema	Dev Team	25-Mar-2026 to 26-Mar-2026
7	Build and test Drawing Reader + Weight Calculator tools	Dev Team	26-Mar-2026 to 28-Mar-2026
8	Run historical quotes through agent and produce comparison reports	Dev + Estimation	28-Mar-2026 to 01-Apr-2026
9	Produce accuracy scorecard and present to management for Phase 2 approval	Dev + Management	02-Apr-2026 to 03-Apr-2026
10	Conduct agent split review — confirm tool count for all agents	Dev + Management	Phase 4 (19-Apr-2026 to 25-Apr-2026)
11	Conduct UAT with estimation team — sign off before go-live	Estimation + Mgmt	Phase 5 (02-May-2026 to 03-May-2026)

Start Date: 25-Mar-2026 | Target Go-Live: 03-May-2026